

A Mini Project-2 Report on SHELF LIFE PREDICTION

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Submitted to the Department of Computer Science & Engineering (AI & ML), GNITS in the partial fulfillment of the academic requirement for the award of B.Tech CSE (AI & ML) under JNTUH, Hyderabad

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Certificate

This is to certify that the Mini Project-2 report on “**SHELF LIFE PREDICTION**” bonafide work carried out by **Srujana Priya (23251A6601)**, **Madhumita C (23251A6638)**, **Khushi Goyal (23251A6647)**, **Sreshta T (23251A6651)** in the partial fulfillment for the award of B. Tech degree in Computer Science & Engineering (AI & ML), G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision.

The results embodied in the Mini project work have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

It is vital to discover the shelf life of the leafy vegetables in order to maintain the safety, quality of the nutritional content in addition to minimizing the wastage of the harvested products. Leafy vegetables such as Spinach hold great value among consumers. They can consistently provide leafy vegetables over the whole harvesting period. On the other hand, the leafy vegetables such as Palak have high demand in the market and high price as well. They are perishable goods that could easily spoil before they reach their end-users because of factors such as wilting, yellowing, physical damages, and the growth of microorganisms. The traditional techniques have been applied to perform the quality control tasks in the case of perishable commodities.

However, the techniques require human intervention in the form of visual examination in addition to professional expertise for performing lab tests on them which were long procedures.

We will design an AI algorithm to predict the shelf life of the harvest using the images of the products to estimate their freshness and remaining shelf life. Our code for developing our AI will involve visual characteristics such as color, texture, structure of the leafy green vegetable to categorize the image file based on its freshness level. In turn, our product will be categorized into one of the three levels namely Fresh, Midlife, and Near Spoilage depending on the freshness level detected from our AI code. This way, we will be able to detect the shelf life of our product in terms of days.

Keywords:

Artificial Intelligence, Shelf Life Prediction, Leafy Vegetables, Image Processing, Convolutional Neural Network, MobileNetV2, Freshness Classification

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1. INTRODUCTION

1.1 Motivation

The use of leafy vegetables is vital because it guarantees the provision of nutrients, vitamins, and minerals to our bodies. However, spinach and other leafy vegetables fall under the category of perishable foods owing to their high water content and delicate nature. After the harvesting process, leafy vegetables wilt, lose color, undergo structural changes, and become prone to bacterial infections. As a result, there are issues related to storage, transportation, and marketing of leafy vegetables.

Presently, the assessment of the quality of leafy vegetables is conducted using visual inspection. The approach is subjective and imprecise. Although laboratory testing produces accurate results, it takes a substantial amount of time to execute. Hence, laboratory testing is impractical because it takes much time to conduct an analysis. The delay can cause food waste due to the early disposal of leafy vegetables or consumption of products that are of poor quality. Hence, food waste is not only a hindrance to economic efficiency but also results in unnecessary food losses.

However, the development of Artificial Intelligence and image processing technologies has enabled the construction of an algorithm that will conduct non-destructive testing of the quality of leafy vegetables and provide estimations regarding the days left before expiration. This is why building an website is the goal of this project.

1.2 Existing Systems

Existing Systems For the problem of shelf life estimation of leafy vegetables, existing methods Tend to rely heavily on traditional approaches. Manual inspection is among the oldest methods that assess the freshness of the vegetables visually by looking at their appearance characteristics, such as their color, texture, firmness, wilting, etc. his approach, however, is very subjective and depends heavily on the person doing the evaluation. It is also hard to scale since the process cannot be applied to mass-produced batches. The lab analysis of the product is another way of evaluating the freshness of vegetables, through microbial and chemical decomposition studies.

While this method offers an accurate estimate of the product's quality, it is both costly slow, taking days or even weeks to return results. Temperatures and humidity levels can be used to estimate the product's shelf life. Another conventional approach would be using conventional

computer vision and machine learning to estimate the product's shelf life. This approach uses hand-coded features, such as color thresholding and texture extraction.

Such an approach is often less accurate compared to the deep learning approach and requires additional procedures to ensure its effectiveness. Thus, the systems used currently are either expensive, subjective, indirect, or not very accurate in general.

Therefore, there is a need to develop an automated, cheap, and AI-based approach to solving the problem.

1.3 Literature Survey

1. Predictive Microbiology-Based Shelf Life Prediction Modelling

Author(s): Scott et al. Date: 1993

Approach: Uses the predictive microbiology approach combined with the modelling technique based on square root and Arrhenius methods. In this context, analysis is carried out in order to determine how the temperature affects microbial growth and spoilage kinetics.

Key Contributions: Laid the groundwork for developing shelf-life predictions through temperature kinetic models which served as a base for further studies in food preservation.

2. Accelerated Shelf Life Testing (ASLT)

Author(s): Zhang et al. Date: 2011

Approach: Accelerated Shelf Life Testing (ASLT), applied using zero-order and first-order kinetic modelling. ASLT helps to reduce time spent on experimentation by exposing samples to extreme conditions of stress.

Key Contributions: The use of ASLT successfully helped to shorten the testing time without affecting prediction accuracy.

3. Hybrid Kinetics – Machine Learning Model Approach

Author(s): Tarlak, F. Date: 2023

Approach: Uses both predictive microbiology and machine learning approach in order to create a hybrid kinetic-ML model.

Key Contributions: Found that combining kinetic models with machine learning techniques results in highly accurate predictions.

4. Shelf Life Prediction Based on Machine Learning Algorithms

Authors: Jubayer et al. Date: 2025

Techniques Used: Includes XGBoost and SVR (Support Vector Regression) The method is aimed at analyzing huge volumes of data and detecting patterns associated with the degradation process in food products.

Important Results: It was established that the use of the XGBoost model yielded maximum precision in predictions compared to other algorithms.

5. Image-based Quality Assessment with CNN

Authors: Infrared Physics & Technology Date: 2024

Techniques Used: Includes CNN (Convolutional Neural Network) The technique is designed for automatic classification of freshness levels according to images obtained from different leafy vegetables.

Important Results: It was discovered that CNN is capable of efficiently classifying freshness stages based on visual characteristics.

6. Shelf Life Prediction Using CNN Approach

Authors: JDIM Date: 2025

Techniques Used: CNN (Convolutional Neural Network) This research is based on applying Convolutional Neural Networks for predicting both freshness level and shelf life duration according to images.

Important Results: It was discovered that CNN is capable of efficient classification and prediction of freshness stages as well as shelf life duration.

1.4 Challenges to existing system

The following difficulties are encountered in shelf-life prediction of leafy vegetables using the current technique. Firstly, there is an issue related to manual inspection. Manual inspection is performed by many people. Inspection is very subjective and relies on perception. It is important

to realize that differences in perception can be significant,so manual inspection is rather subjective. What is more, the technique in question cannot be implemented for a large number of items because it takes much time. Thus,it works well when only a few items are inspected. Chemical and laboratory testing can be considered a fairly reliable method. However, it is rather expensive and requires much time. Chemical testing requires specific equipment as well as qualified personnel, so this method does not work well in farms and retail stores. The analysis of humidity and temperature provides just indirect information about the freshness of vegetables, while physical and visual characteristics are not checked. In addition,special equipment is required

Traditional computer vision methods imply that the features have to be Selected manually (e.g., color and texture analysis). The problem with such approaches is related to their inability to operate under changing lighting conditions. Hence a new system is to be designed to overcome these problems.

1.5 Proposed system

The proposed system is a shelf life prediction system that is driven by artificial intelligence and is used to predict the shelf life of leafy vegetables depending on the level of freshness using image analysis. Images of leafy vegetables, including spinach, mint, coriander, and cabbage, are taken using an RGB camera to be classified in accordance with the level of freshness.

The pre-processing steps include resizing, normalizing, and augmenting of data to increase quality of inputs. The classification process utilizes Deep Learning and Transfer Learning algorithms together with MobileNetV2 network to identify critical features, such as color degradation and texture of leafy vegetables. The images are categorized as Fresh, Mild (mid-life), and Rotten (near spoilage). Days of storage can be calculated for each category. The proposed system enables fast non-destructive testing of the freshness stage of the product without requiring any lab analysis or sensors.

1.6 Objectives

The critical elements that make up this study include:

- Creation of novel deep learning approach to estimate the shelf-life of leafy vegetables by analyzing their images
- Implementation of a CNN architecture that utilizes the MobileNetV2 framework to predictwhether the leafy vegetables belong to any of the three categories: fresh, mid-life, or

rotten.

- Pre-processing of the input images through a number of image transformations such as scaling, normalization, and data augmentation.
- Analysis of visual cues via imaging technologies in RGB format that detect the presence of certain characteristics in leafy vegetables including their color pattern, texture, or wilting state.
- Estimation of the predicted shelf-life measured in days based on the results of freshness analysis.
- Verification of the effectiveness of the design through leafy vegetable prototyping, i.e. spinach.

1.7 Methodology

A number of structured steps are applied within the framework of the methodology for predicting the shelf life of newly developed system, including data gathering and Prediction. At first, images of leafy vegetables such as spinach, mint, coriander, cabbage are grouped into three clusters on the basis of their freshness (Fresh, Mild (Midlife), Rotten). At second, a procedure of train-test-validation was implemented for more precise measurement of the model's performance. In terms of preprocessing, all images have been transformed in order to become equal in size to those of 224 x 224 pixels, as required by the Base model. Apart from that, normalization of the pictures is also performed during this process for better results.

Speaking about model creation, Transfer Learning is used through the implementation of pre-trained model – MobileNetV2. The Base model was frozen, while new layers were introduced to solve the task of classification (Global Average Pooling, Dropout (rate = 0.5), Dense layer, and Softmax activation). Optimization and validation of the model took place by using Adam optimizer and categorical cross-entropy loss function. Early stopping was used. Finally, class prediction and remaining shelf life calculation in days were made by the model.

1.8 Hardware & Software requirements

Hardware Requirements:

Computer/Laptop with a minimum of 8GB RAM,

Processor- Intel Core i5 and above (or equivalents),

Camera/phone camera to take the photo,

Optionally Graphics Processing Unit (Recommended),

Connection with the internet to conduct research work and to upload on cloud platform if required.

Software Requirements:

Operating system can be one among Windows/Linux/MAC OS,
Programming language – Python,
Open CV- to manipulate the image,
NumPy/Pandas for data manipulation,
Matplotlib/seaborn to draw the visualization graph,
Scikit learn library to determine the accuracy of the model,
TensorFlow/keras or Pytorch library to develop deep learning models,
Jupyter Notebook/Visual Studio/Google Collab,
Dataset for leafy vegetable classification

1.9 Organization of the project

The Shelf Life Prediction project consists of several stages that make up a development lifecycle from collecting raw data to the final testing. Initially, there is a stage associated with the collection of data. At this stage, images of the randomly selected leafy vegetables are collected and classified according to their level of freshness. After that, there is a stage called image preprocessing. Here, images are preprocessed, normalized, and augmented to have standardized inputs.

Then follows the creation of a machine learning model using Transfer Learning algorithm and MobileNetV2 architecture. The created model classifies the level of freshness of an image by applying the technique of dropout regularization. After finishing the training process, the model is ready for prediction, which gives the number of days left for the shelf life.

Finally, there is a testing phase, during which the quality of the predictions will be evaluated.

2. PROPOSED SYSTEM

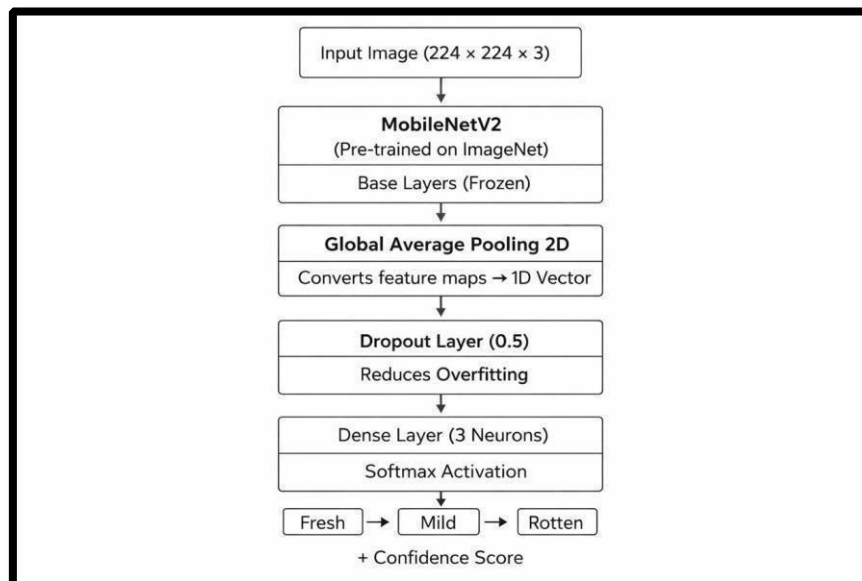
2.1 Architecture

Architecture of the Proposed Shelf Life Prediction system. The Proposed Shelf-life Prediction system uses deep learning techniques by leveraging transfer learning from the MobileNetV2 Convolutional Neural Network model.

Firstly, an image of leafy vegetables is captured and classified based on the condition of its freshness. Initially, an image of leafy vegetables having a dimension of $224 \times 224 \times 3$ is obtained wherein the dimensions represent the RGB image of the leafy vegetables. Thereafter, the image is used as input data for the pre-trained MobileNetV2 Convolutional Neural Network.

The pre-trained base layers of the MobileNetV2 neural network are kept frozen in order to utilize the learned features such as edges, texture, colors, etc. Next comes the Global Average Pooling 2D layer in which high-dimensional feature maps are converted into 1-D vectors. Afterwards, a dropout layer with a probability of 0.5 is added. To avoid overfitting, the role of this layer is to drop out a few neurons randomly. After this, the output will be sent to a dense layer that has three neurons in it. Then, the Softmax activation function is employed to assign probabilities to each class. Finally, the output stage of classification will have three classes for freshness - fresh, mild, and rotten. Along with the classification outcome, the model provides the probability of the assigned class.

Figure 2.1.1 Architecture flow of the model



2.2 Module level design

The Shelf Life Prediction Application is made up of several modules that assist in predicting the freshness of the leafy vegetables as well as predicting their shelf life.

2.1.1 Image input module

Images are obtained through the use of the input image module. The input image module helps capture images using the camera or any other existing dataset. Once images have been obtained, there arises a need to resize the images to 224 x 224 x 3 for deep learning.

2.1.2 Feature extraction module

The input image is transformed into something meaningful through the use of the feature extraction module with the MobileNetV2 pre-trained model. The feature extraction module freezes the base layers of the input image to obtain features.

2.1.3 Feature processing module

The feature maps that are extracted from the image data are processed using a Global Average Pooling 2D layer. The purpose of this process is to convert the feature maps into a single dimensional feature vector. This approach simplifies the computation, ensuring only the important features are considered for classification.

2.1.4 Regularization module

A Dropout layer with a probability rate of 0.5 is used for preventing the issue of overfitting. In this module, some neurons are deactivated during training to make the model more generalizable.

2.1.5 Classification module

In the classification module, a Dense layer having three neurons with Softmax Activation Function is used to classify the image into one of the three stages of fruit freshness.

2.1.6 Output module

The output module outputs the result by showing the freshness category together with its confidence level. Using the category, the shelf life left for the leafy vegetable can be estimated.

3. IMPLEMENTATION AND TESTING

3.1 Implementation

The Shelf Life Prediction system is designed through a highly structured implementation strategy that entails the following stages; preparation, pre-processing, development, training and prediction models. The process of implementing this system involves employing Python Programming Language and deep neural networks.

3.1.1 Dataset preparation

A selection of some photos depicting common leafy vegetables such as spinach, mint, and coriander is included in this project. The different aged cabbages that have varying levels of decomposition are grouped into either fresh, mild or old. These photos are sorted out based on their categories by placing them in various folders for the generation of both the training and validation data sets.

3.1.2 Data preprocessing

All the images used in this project are adjusted to the size of 224 x 224 pixels to meet the input size of the classification model implemented. Implementation of MobileNetV2 network. Pixel Normalization is conducted within the range of 0 - 1. Two Data Augmentation strategies; image rotation and image flipping are applied. Image Rescaling and Image Zooming techniques are also applied during the training phase.

3.1.3 Model development

Transfer Learning was performed using the pre-trained model MobileNetV2. These models were trained with the use of data from ImageNet. The base layers of this model were freezed in order to preserve its trained features. Features. Additional layers were added such as Global Average Pooling to reduce the feature space dimension, Dropout (0.5) to prevent overfitting, and Dense layer. The output layer had three neurons with SoftMax activation for the classes “Fresh”, “Mild”, and “Rotten”.

3.1.4 Model training

The training of the model was done by the Adam optimizer and categorical cross entropy loss. The entropy loss function was used during the training of the network models in different epochs. Accuracy on validation set. To prevent overfitting, the early stopping method was used.

3.1.5 Prediction and Shelf Life estimation

After the training stage of the deep learning model has been done, the model gains the ability to predict the input data as being one among the three classes of data in the model. Prediction of the input data coordinates can be made by running the frame captured by the camera through the model. The class in consideration is the one used in this experiment. Prediction of the input images for each of the three classes was made depending on how accurate the model prediction would be. Based on the mapping stage, prediction of the class could estimate the days left for shelf life.

3.2 Testing

Test of the proposed prediction system would make the effectiveness of the system known. The next chapter will outline briefly the system of prediction that would be developed. To test the generalization ability of the dataset, images have been taken which belong to the test data set that has not been used previously. The testing of the model was conducted by providing test input images to the model and then those images were tested through our MobileNetV2 model that has been trained for this purpose.

The method that has been employed to measure the quality of leafy vegetables through visual feature extraction has proven quite effective since a very high accuracy level was obtained. Spinach has been considered in our model.

Furthermore, we have also tested shelf-life estimation module under room temperature. Unlike conventional approaches which rely on manual annotation techniques, predictions made by the Caffe Model in relation to the level of image have been proven to be more efficient than manual inspections. The achieved results have shown the efficiency of the implemented model for detection. Dimensionality can be considered an effective parameter to estimate the freshness of food and other consumer goods.

3.3 Description of Technologies / Tools used

Frontend:

Fast ai Libraries: The fast ai libraries like DataBlock, Device, Pipeline and Learner are used to perform this process. This tool is used to run the Python code. There are some functionalities provided by this code through which users can upload their images and some suggestions can be provided on the basis of analysis of these images by the model. The model will run and its working will also be visualized here. This code will help to implement this process on the GPU for faster output.

Backend Python:

The programming language that we used to create this system is Python and it has very simple syntax using different libraries and image processing.

TensorFlow / Keras:

Tensorflow and Keras were both used for implementing this model because it is a Deep Learning model.

MobileNetV2:

MobileNetV2 is a lightweight CNN-based Deep belief networks architecture which can be applied to the process of feature extraction and image classification.

OpenCV:

OpenCV is a library that contains easy implementations of numerous image processing algorithms. For instance: reading/writing images, basic scaling of images. These files are opened by the algorithm, and several preprocessing actions are performed on the model.

NumPy:

This library can be applied to various purposes like numerous numerical and array manipulations. Function processing of image data and mathematical operations while modeling.

Matplotlib:

The Python Matplotlib library is used to visualize the results of model training such as loss on training and validation, error test accuracy, and graph of losses.

3.4 Modules description

1. Image input module:

Image Input: Users may upload an image of the leafy vegetable. **Input Processing:** Once you upload the image, the software processes the image.

2. Image preprocessing module image resizing:

The input image is resized to 224 in width or height to act as the dimension of the input of MobileNetV2.

Image Normalization: Normalization of pixels ensures uniformity of the data to be inputted in the deep learning model.

Processing the Image to Array: After the above process, the output is converted to an array which acts as the input of the model for prediction.

3. Feature Extraction Module MobileNetV2 Model:

The pre-trained convolutional neural network MobileNetV2 model is used to extract the visual features of the input image. **Feature Extraction:** Neural Network recognizes the changes in color, texture, and wilting of leaves

The pre-trained convolutional neural network MobileNetV2 model is used to extract the visual features of the input image. **Feature Extraction:** Neural Network recognizes the changes in color, texture, and wilting of leaves.

4. Feature Processing Module Global Average Pooling:

Dimension Reduction: The high dimensional features extracted from the deep learning model are reduced to one feature vector.

Dropout Layer: Dropout technique applied during training to minimize the risk of overfitting.

5. Classification Module

Dense Layer: Dense layer embedding the extracted features. Initially, feature extraction will be performed followed by its application in a dense layer with three channels.

Softmax Function: Three scores will be obtained through this function, which are vital when classifying three categories of freshness level; they include fresh, medium, and rotten.

6. Result Display Module:

Classification Output: Results from the classification process of the leafy vegetable freshness levels.

Probability: Score denoting the probability level of the output prediction result

7.Feature Extraction Module MobileNetV2 Model:

Pre-trained MobileNetV2: Trained model used for feature detection.

Feature Detection: Identification of visual features such as color, texture, and vegetable wilting.

8. Feature Processing Module Global Average Pooling:

Global Average Pooling: Simplifies the model by transforming the feature maps into a single vector layer.

Dropout Layer: The rate of 0.5 is utilized to avoid overfitting.

9. Classification Module

Dense Layer: The processed features are forwarded features are forwarded to the dense layer of three neurons.

SoftMax Activation: Gives probabilistic output for three levels of freshness – Fresh, Mild, and Rotten.

10. Result Display Module

Prediction Output: Prediction result for the level of freshness of leafy vegetables.

Confidence Score: Represents probabilistic values for prediction accuracy

4. RESULTS AND DESCRIPTION

The developed Shelf Life Prediction system incorporated the use of deep learning with the application of the MobileNetV2 model. The model utilized a dataset of images of leafy vegetables that were separated into three categories of class labels: Fresh, Mild (Mid-life), and Rotten. This was done using the Google Colab development environment that utilized all the necessary machine learning libraries.

In the training process, the deep learning model learned how to analyze various visual features in relation to freshness. The learning process entailed the processes of image pre-processing, feature extraction using MobileNetV2 model and finally image classification using the fully connected layer.

The trained deep learning model could be able to predict the freshness stage of the leafy vegetable through the input images. Upon uploading a new image, the system evaluated its features and gave the output of freshness category with confidence score. From the results, it was evident that the deep learning model could effectively classify the freshness stages of the vegetable.

The output of the system can therefore be regarded as an effective means of evaluating the freshness state of leafy vegetables.

Therefore, it should be stated that thanks to this method of classifying products, it is now possible to estimate their freshness level and predict the amount of shelf life time left. Also, this system may potentially be improved by adding additional types of vegetables to classify their freshness.

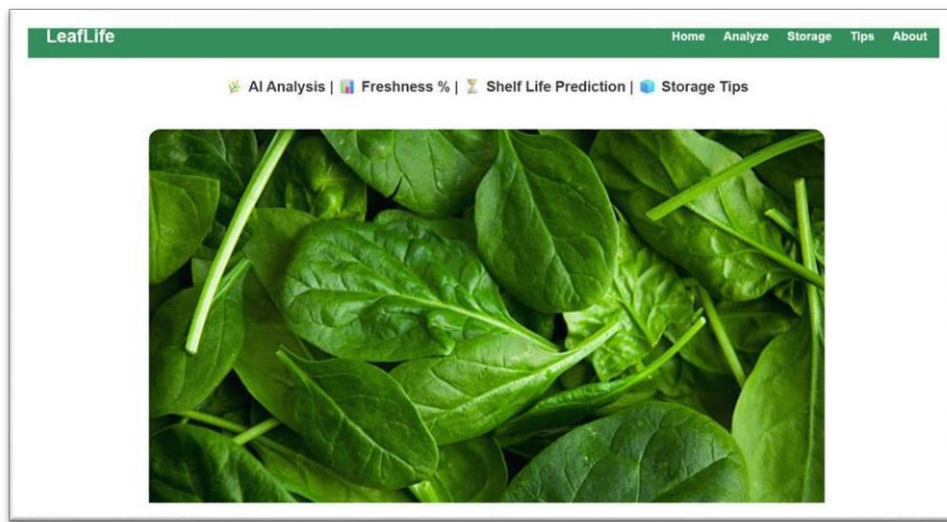


Figure 4.1 Spinach Image Input on LeafLife Platform for Shelf Life Prediction

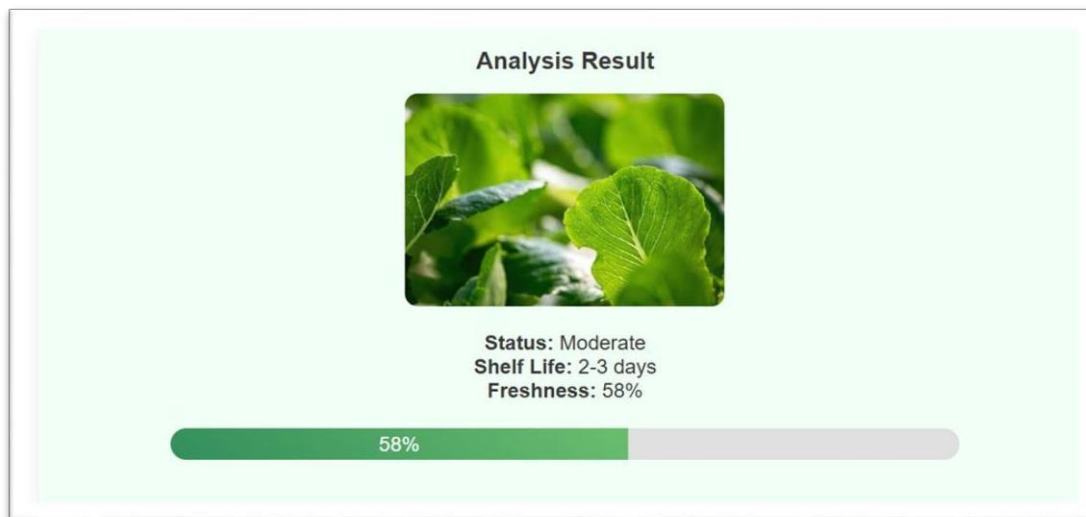


Figure 4.2 Model Prediction Result with Freshness Percentage and Shelf Life Estimation



Figure 4.3 LeafLife Interface Displaying Freshness Tips for Spoiled Produce

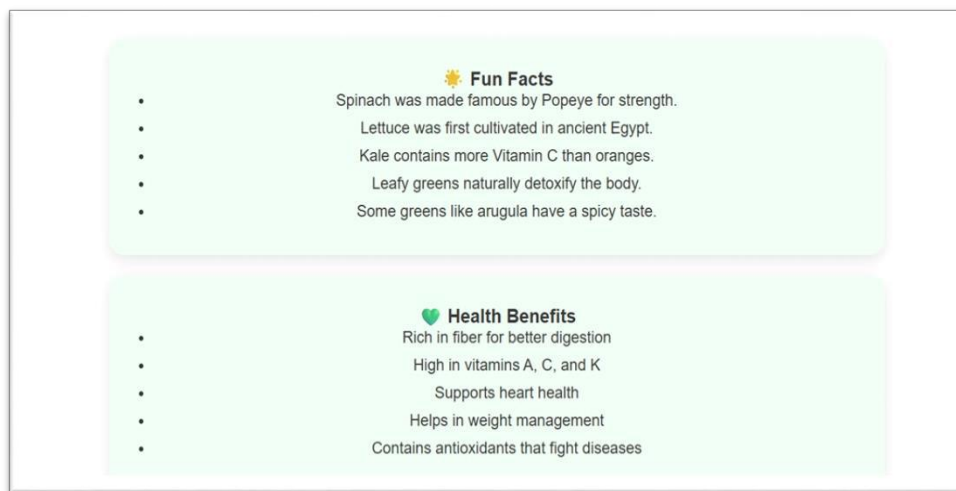


Figure 4.4 LeafLife Interface Displaying Fun Facts and Health Benefits of Leafy Greens



Storage Guide for Leafy Vegetables



Fresh Leaves (70% - 100%)

- Store in refrigerator immediately
- Wrap leaves in paper towel to absorb moisture
- Keep inside airtight container
- Do not wash before storing
- Keep in vegetable crisper drawer
- Avoid storing near fruits like banana and apple
- Ideal temperature: 1°C – 4°C

Figure 4.5 Leaf Life Storage Guide for Fresh Leafy Vegetables

5. CONCLUSION

The Shelf-Life Prediction System utilizes a set of machine learning and image processing methods used for solving the issue of determining the freshness level of leafy greens. As for the key component of the system, it is the convolutional neural network. It is based on one particular type of neural network called MobileNetV2, which is specially developed for image recognition tasks. The process itself is rather simple and involves the analysis of the degree of the freshness factors present. In other words, there will be estimated the presence of Fresh, Mild, and Average degrees of freshness.

For detecting freshness level, it is critical to analyze a variety of visual attributes found in the vegetables. Such attributes may include changes in the color, which is an indicator of vegetable spoilage, texture changes that imply a loss of firmness and blurring or fading out that show deterioration. In addition, the system is also able to recognize patterns of vegetables' wilting, which include, for instance, vegetable leaves becoming soft and droopy, etc., which is one of the main attributes of a less fresh vegetable. Thus, through training on specific patterns, the CNN is able to accurately make forecasts upon receiving new images.

The major advantage of such system lies in its ability to work fast and efficiently. As such, the system allows minimizing the use of human resources, avoiding human error and subjectivity and, therefore, working fast. In this respect, such a system would be particularly helpful when there is a necessity to inspect lots of vegetables, such as in markets or in warehouses.

Moreover, such systems are also expected to become helpful when it comes to inspection of different food products other than vegetables. That is why, the presented approach may be considered useful for food quality control. With the help of early identification of food decay, the technology plays a huge role in food preservation, a problem that is very crucial globally.

6. FUTURE WORK

Nevertheless, there are additional possibilities for enhancing the prediction process within the current system. Firstly, it is possible to expand the database used for training. An increased amount of images will facilitate a higher effectiveness of the system because it will have to deal with vast amounts of information related to vegetables, including illumination, backgrounds, and other aspects. In such a case, the model will be more resistant to environmental changes, which implies that it will be able to identify fresh products regardless of conditions.

In addition, the system can be trained to assess different kinds of vegetables simultaneously. There are multiple benefits to this approach. For example, the scope of application of the system will be extended as it will be capable of examining various vegetables by their individual characteristics. Moreover, the system can be integrated into a mobile application aimed at determining the freshness of vegetables.

Furthermore, the suggested framework can also be combined with the IoT-based storage systems to check the freshness of vegetables through IoT sensors, and all required variables such as temperature and humidity would be measured; then the acquired information could be evaluated by the proposed predictive framework based on the characteristics of the vegetable. This would make sure that the storage process is always under optimal conditions and prevent any chances of decaying the vegetables.

Additionally, another alternative for increasing the efficiency of the model might involve trying out other machine learning models such as deep learning algorithms. This would increase the efficiency and accuracy of the predictive system that has been developed. In conclusion, with continuous improvements in the mentioned approaches and techniques, the shelf life of the prediction model can be greatly enhanced.

7. REFERENCES

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8.GLOSSARY

Artificial Intelligence (AI):

AI refers to a field within computer science devoted to the creation of intelligent machines capable of doing tasks typically requiring human intelligence.

Machine Learning (ML):

ML refers to a sub-discipline of AI devoted to making computer systems learn from data and discover patterns within the data without explicit programming.

Deep Learning:

Deep Learning is another sub-discipline of machine learning that entails creating artificial neural networks comprising several layers to analyze complex data and discover complicated patterns within data.

Convolutional Neural Network (CNN):

CNN is a specific form of deep learning model that can be applied effectively in analyzing images and discovering patterns within the images. In this case, the network automatically discovers the important features.

MobileNetV2:

MobileNetV2 is an efficient deep learning neural network architecture that can be used for image classification.

Image Processing:

Image processing is defined by the procedures involved in improving the quality of the image or extracting information from the image.

Dataset:

A dataset can be described as the data used in training the machine learning algorithm.

Image Preprocessing:

Image preprocessing is one of the processes involved in preparing the images for use in the machine learning algorithm. The image preprocessing process includes image normalization, image resizing, and image augmentation.

Freshness Classification:

The categorization of the state of the vegetable as either fresh, moderately fresh, or rotten is referred to as freshness classification.

Shelf Life:

The period within which the product can still be safely consumed without any degradation is known as the shelf life.

9.APPENDIX

Transfer Learning:

In transfer learning, a deep learning model that is pre-trained previously is used again for some other similar but distinct task. Rather than creating a new deep learning model by using training approach, pre-trained deep learning model's knowledge (weights obtained through the process of training with a huge dataset such as ImageNet) is shifted to some other task.

MobileNetV2 Architecture:

MobileNetV2 is an example of lightweight deep learning architecture which can easily be trained for classifying images. Depth-wise separable convolution and inverted residual blocks have been used in this model so as to cut back computation power required.

Feature Extraction:

Feature extraction refers to extracting features from images. Features may include color, shape, edges, etc.

Global Average Pooling (GAP):

Global Average Pooling method is used to bring down each of the feature maps to one single value by applying the averaging method. GAP makes the model have less number of parameters making it stable.

Dropout Technique:

The Dropout method is a form of regularization method where the neurons will be randomly ignored while training the model. In this method, the model will not focus too much on specific neurons, hence improving generalization.

SoftMax Function:

The Softmax activation function is applied in the last layer of the neural network model so as to make sure the output becomes probabilities. The class with the highest value will get the largest probability.

Data Augmentation:

Data augmentation method is used to increase the diversity of the training data through application of rotations, flips, zoom, colors and many more.

Adam Optimizer:

This, which is also referred to as an adaptive moment estimation optimizer, is an algorithm used in optimizing model parameters when training a neural network. Adam optimizer was selected due to its capability of being able to adaptively adjust the learning rate as well as use the concept of momentum.

Categorical Cross Entropy Loss Function:

It is the function applied in calculating the loss in multiclass classification problems.

